

MODEX GEN AI - WEEK 2 SUMMARY :

week2-summary.md

WHAT WAS BUILT:

1. In week 2, I worked on Different Practical activities that Helped me understand How LLM application are Designed, Validated, Compared, and evaluated.
2. I started By working with Multiple models such as Gemini, Claude and Llama.
3. I compared their Responses using the Same Prompts and analysed the Different in Quality, Response style, latency and Prompt Portability, this Helped me understand that the same Prompt can Produce Different Result across Different models.
4. I also worked on structured output generation using JSON Schemas and Pydantic Validation. Instead of accepting Free-Form text Responses, I learned How to Force the model to Return structured output :

5. I Built a compliance extraction utility that extracts Rule ID's, compliance status, evidence quotes, confidence scores, and note from inspection reports.
6. To improve Reliability, I Implemented Validation checks and Retry logic. This Helped Handle cases where the model generated Invalid outputs.
7. Another major task was Building an evaluation Framework for the compliance extraction system.
8. I created a golden Dataset containing multiple test cases with expected output and Developed an evaluation compare the model to output (v/s) the expected Result.
9. Using the evaluation Results, I Performed Failure analyse; Identified common Failure Patterns, categorized them into Different Failure types.

10. Throughout the week, I also created supporting documentation, comparison reports, observation notes, evaluation reports and iteration summaries to document the learning process and track improvements across different stages of development.

WHAT WAS LEARNED:

1. During week 2, I learned that building a GenAI application involves much more than simply calling a model and getting a response.
2. The quality of the output depends on prompt design, structured outputs, validation evaluation and continuous improvements.
3. One of the key things I learned was prompt variability. By comparing Geminii, Claude and llama, I saw that the same prompt can produce different responses depending on the model. Each model has its own strengths, response style and performance.

4. I also learned how structured output makes AI applications more reliable. Using Pydantic schemas helped me understand how to define expected output formats and validate model responses before using them in an application.

5. Another important learning was the difference between JSON mode and schema validation. JSON mode helps generate structured JSON responses, but it does not guarantee that the fields (or) data types are correct. Pydantic validation helps verify that the output follows the expected schema.

6. I learned why retry logic is useful when working with LLMs. If the model generates an invalid response, retries provide another opportunity to produce a valid output instead of immediately failing.

7. I learned from this week:

1. How to create a golden set
2. Build an evaluation harness
3. Analyse failures.

4. Improve Prompt using measurable result.
3. Finally week 2 Helped me understand How Gen AI systems are Designed, Validated, evaluated, and Improved in a structured and Reliable way.

BIGGEST CHALLENGE:

1. The Biggest challenge for me During week 2 was making the model output more Reliable and consistent.
2. While working on the compliance extraction Project, I noticed that the model Did not always Return the expected output.
3. understanding schema Validation, Handling Validation errors, and Implementing Retry logic took time and Required multiple tests.
4. Another challenge was the evaluation Process, creating a golden set, analyzing Failed cases, Identifying the Reason for Failures, and Improving the Prompt Based on the Result was a new experience for me

5. This week Helped me understand that Building a GenAI application is not only about getting Responses from a model But also about Validating evaluating and continuously improving the system.

BIGGEST IMPROVEMENT:

1. The Biggest Improvement for me During a week was learning How to measure and Improve the Performance of an AI system.
2. At First, I used to check outputs manually and Decide whether they looked correct (or) not.
3. After working on the evaluation Harness and golden set, I learned How to use scores and evaluation Result to measure Performance more accurately.
4. By analysing Failed cases and Improving the Prompt through multiple iterations, I was able to Improve the overall evaluation score.

5. This Help me understand How Important testing and evaluation are in GenAI application.
6. I also improved my understanding of structured outputs, Pydantic validation, Retry logic, model comparison and evaluation methods.
7. compared to the Beginning of the week, I now have a much Better understanding of How Reliable GenAI system are Built and Improved.

TAKAWAYS:

1. The main takeaway From week 2 Is that Building a GenAI application involves much more than simply generating Responses From a model.
2. I learned that Prompt Design Plays an Important Role in the Quality of the output and that Different models can Respond Differently even when the same Prompt Is used.
3. I also learned the Importance of structured output, schema validation and Retry mechanisms in making AI system more Reliable.

4. Another key learning was evaluating (or) evaluation, creating a golden set, measuring Performance, analysing Failure and Improving Prompts Helped me understand How to make Decisions Based on Data Instead of assumptions.

5. overall, week 2 Helped me understand How GenAI Systems are Built, Validated, evaluated and Improved in a more structured and Practical way.